INCM PROPOSAL Topic 27: Evidence Accumulation

Modeling Simple Decisions & Speed-Accuracy Trade-off

1D Drift Diffusion Model analysis across task difficulties and dynamic decision thresholds

Continuous Accumulation

Brain gathers noisy sensory evidence over time rather than deciding from a single snapshot.

Task Difficulty Patterns

Easy tasks trigger rapid, accurate choices; hard or faint stimuli cause delays and higher error rates.

Choice & Thresholds (SAT)

Accumulation explains choices and reaction times; decision thresholds dictate the speed-accuracy balance.

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PROJECT SCOPE & RESEARCH OBJECTIVES

Detailed Problem Statement

Simulator Objective

Build a trial-by-trial Drift Diffusion Model (DDM) simulation in Python to systematically analyze decision-making mechanics across stimulus contrasts and decision boundaries.

1. TASK DIFFICULTY

Vary contrast (0%–100%) to reproduce psychometric and chronometric curves.

2. SPEED VS. ACCURACY

Adjust thresholds from low (rushed) to high (cautious) to map SAT curves.

3. COLLAPSING BOUNDARIES

Apply urgency decay to fix unrealistically long RTs on zero-contrast trials.

4. MODEL ASSUMPTIONS

Evaluate biological realism of constant caution, noise, and motor delays.

QUESTION 1

How well does noisy evidence accumulation match empirical accuracy and RT curves?

QUESTION 2

Can collapsing boundaries solve long-tail reaction times on zero-contrast trials?

QUESTION 3

What do core model assumptions tell us about neural uncertainty and deliberation costs?

QUANTITATIVE ARCHITECTURE

Methodology & Simulation Protocol

1D Diffusion Stochastic Process

```
x(t+Δt) = x(t) + v(c)Δt + σ√Δt ξ(t)
v(c) = k · c_i | ξ(t) ~ N(0, 1)
RT = T_d + t_0
```

Collapsing Boundary Extension

$\theta(t) = \pm a \cdot \exp(-(t / \lambda)^{\gamma})$

Euler–Maruyama Integration
1ms step size with 3–second maximum duration cutoff.

Parameter Calibration
Nelder-Mead simplex fitting on choice fraction and RT distributions.

Model Evaluation Metrics
Psychometric, chronometric, and BIC goodness-of-fit comparison.

1 ms	6 Levels	BIC / Likelihood
Simulation Step (Δt)	Visual Contrast Range	Boundary Selection Metric

FOUNDATIONS & BENCHMARKS

Data Repositories & Key References

Data Repositories

IBL Mouse Visual 2AFC Dataset
International Brain Laboratory (2021) public behavioral repository.

- **Contrast Levels:** 0%, 6.25%, 12.5%, 25%, 50%, 100%
- **Trial Filtering:** Exclude RT < 100ms and > 2.5s
- **Synthetic Data:** Known parameter recovery verification

Key Literature

Ratcliff & McKoon (2008) — DDM theory and two-choice decision dynamics.

Bogacz et al. (2006) — Mathematical SPRT equivalence and optimality.

Drugowitsch et al. (2012) — Evidence accumulation costs and urgency signals.

IBL (2021) — Standardized mouse decision-making dataset in eLife.

Framework: Python 3 with NumPy, SciPy (Nelder-Mead), and Matplotlib

EXECUTION PLAN

Project Timeline & Milestones

WEEK 1

Simulator Development

Implement Python Euler-Maruyama simulator and test synthetic parameter recovery.

WEEK 2

Data Pipeline & Fitting

Process IBL mouse dataset and fit fixed-boundary DDM via Nelder-Mead optimization.

WEEK 3

Collapsing Boundary

Implement exponential threshold decay and perform BIC model comparison.

WEEK 4

Evaluation & Report

Generate psychometric & chronometric curves and finalize project deliverable.